

EMLEY MOOR, United Kingdom

WMO: 033460

Lat: 53.612N

Lon: 1.667W

Elev: 876

StdP: 14.24

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 26.3	(c) 28.1	(d) 22.4	(e) 17.4	(f) 30.5	(g) 24.4	(h) 19.1	(i) 31.4	(j) 63.4	(k) 42.2	(l) 57.6	(m) 40.5	(n) 9.6	(o) 300	(p) 0.690

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	74.9	63.5	71.3	62.0	68.3	60.3	65.0	72.0	63.1	69.2	61.4	66.9	9.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.3	86.7	68.1	60.6	81.5	65.9	59.0	77.0	64.0	30.5	72.2	29.0	69.6	27.8	66.8	73.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB
(a) 51.3	(b) 41.2	(c) 31.2	(d) 23.4	(e) 80.6	(f) 2.8	(g) 3.6	(h) 21.4	(i) 83.2	(j) 19.8	(k) 85.3	(l) 18.3	(m) 87.3	(n) 16.2	(o) 89.9	(p) 74.7	
(a) 51.3	(b) 41.2	(c) 31.2	(d) 22.8	(e) 68.7	(f) 2.7	(g) 2.3	(h) 20.8	(i) 70.3	(j) 19.3	(k) 71.7	(l) 17.7	(m) 73.0	(n) 15.8	(o) 74.7	(p) 74.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.0	38.5	38.7	41.3	44.7	50.6	55.5	59.6	59.2	55.0	49.5	43.4	39.7
	DBStd	9.17	5.34	5.67	5.43	5.33	5.44	4.85	4.72	4.30	4.32	4.90	5.06	5.67
	HDD50	1780	356	318	273	172	58	7	0	0	6	67	203	320
	HDD65	6215	821	737	734	608	447	287	180	188	302	480	647	784
	CDD50	1068	0	2	3	15	76	172	297	287	156	53	6	1
	CDD65	26	0	0	0	0	0	3	12	9	2	0	0	0
	CDH74	131	0	0	0	0	0	22	63	43	2	1	0	0
	CDH80	8	0	0	0	0	0	1	5	2	0	0	0	0
Wind	WSAvg	13.0	16.4	15.5	14.1	11.8	11.5	10.6	10.1	10.7	10.4	13.6	14.2	17.1
Precipitation	PrecAvg	36.9	3.5	2.7	2.5	2.6	2.4	3.1	2.8	3.1	3.0	3.6	3.6	3.9
	PrecMax	46.6	7.5	6.1	4.4	5.4	4.5	10.6	5.7	7.6	6.0	6.7	7.1	7.6
	PrecMin	27.6	0.7	0.4	0.3	0.5	0.6	0.9	1.2	0.9	1.3	1.3	1.8	1.9
	PrecStd	5.0	1.7	1.3	1.1	1.3	0.9	2.2	1.3	1.7	1.4	1.5	1.6	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.9	55.9	57.6	64.9	71.7	77.3	79.2	79.2	72.7	65.5	57.0	54.8
		MCWB	50.0	50.0	49.9	55.4	61.0	63.4	64.9	64.9	63.3	58.8	54.8	52.4
	2%	DB	50.2	51.3	53.6	59.1	67.1	71.0	75.4	73.2	67.7	61.0	54.8	52.3
		MCWB	47.8	47.6	48.7	51.5	57.6	60.8	63.6	63.3	60.3	57.0	52.7	50.4
	5%	DB	48.1	48.8	51.4	55.6	62.9	67.1	71.9	69.3	64.3	58.5	52.7	49.9
		MCWB	45.8	45.9	47.4	49.6	55.2	58.7	62.3	61.6	58.4	55.4	50.6	48.1
	10%	DB	46.3	46.8	49.2	53.0	59.5	64.0	68.4	66.6	61.6	56.7	50.7	47.9
		MCWB	44.3	44.2	45.8	47.7	53.2	56.8	60.6	60.1	56.8	53.9	48.9	46.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.6	51.1	52.1	56.1	61.8	65.7	67.5	67.4	65.5	60.1	55.1	53.1
		MCDB	52.1	53.7	55.0	63.0	68.7	74.8	75.8	74.7	70.6	63.8	56.4	54.4
	2%	WB	48.3	48.6	50.0	52.7	58.7	62.1	65.0	64.7	61.7	57.8	53.1	50.5
		MCDB	49.9	51.1	52.8	57.8	65.2	68.5	72.5	70.6	66.1	60.4	54.5	52.1
	5%	WB	46.3	46.4	48.0	50.4	56.2	59.8	63.1	62.7	59.3	56.1	50.9	48.6
		MCDB	47.7	48.5	50.6	54.8	61.8	65.6	69.7	68.4	62.8	58.4	52.3	49.9
	10%	WB	44.6	44.5	46.1	48.3	54.0	58.0	61.2	61.1	57.7	54.2	49.2	46.3
		MCDB	46.2	46.6	49.0	52.2	58.6	63.0	67.3	65.6	60.9	56.3	50.5	47.8
Mean Daily Temperature Range	5% DB	MDBR	6.6	7.5	9.3	11.2	12.4	12.3	12.6	11.6	10.4	8.5	7.0	6.7
		MCDBR	9.2	9.4	11.8	14.5	17.8	17.8	18.6	16.1	14.1	10.3	8.8	9.0
		MCWBR	9.1	8.3	8.3	8.9	10.7	9.9	9.0	8.1	8.6	7.6	8.1	8.9
	5% WB	MCDBR	8.9	9.1	10.9	13.9	16.4	16.0	16.5	14.3	12.2	10.0	8.6	9.2
		MCWBR	8.9	8.4	8.7	9.4	10.7	10.2	9.4	8.3	8.7	8.2	8.2	9.2
Clear-Sky Solar Irradiance	taub	0.314	0.328	0.357	0.382	0.384	0.385	0.385	0.372	0.355	0.338	0.313	0.301	
	taud	2.345	2.399	2.322	2.267	2.301	2.318	2.318	2.373	2.430	2.456	2.407	2.315	
	Ebn at Noon	207	245	260	268	274	274	272	269	259	238	207	186	
	Edh at Noon	18	24	31	38	38	38	37	34	28	22	17	16	
All-Sky Solar Radiation	RadAvg	206	420	765	1188	1500	1549	1501	1236	903	511	254	163	
	RadStd	21	48	88	136	100	144	153	81	80	44	28	17	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (70 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page